

Core Signatures and Inversions

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Abstract

Core signatures retain the entries of a truncated path signature indexed by Lyndon words. Classical shuffle-algebra results show that these entries are polynomial coordinates: they carry the same information as the truncated signature, with the dimension of the log signature and without a Lie-basis projection. We give an explicit elimination procedure and substitution rules through level five, and study recovery of time-augmented paths from these coordinates. For two linear segments we characterise exact recovery by a compatibility condition and an admissible breakpoint. The coordinates containing one value increment are moments of that increment; this gives exact polynomial recovery and a finite-level stability estimate on bounded polynomial families. A rank condition gives local recovery on smooth parameterised families. We also calculate the covariance of the moment coordinates under additive Brownian-bridge noise. Two counterexamples delimit the theory: undamped factorial weighting can make the all-level residual infinite, and signature depth cannot remove the approximation barrier of a fixed piecewise-linear class. A damped all-level metric recovers the uniform topology on an anchored class with a common Lipschitz bound, without asserting Sobolev norm equivalence. A Python implementation and two interest-rate curves illustrate the distinction between exact recovery, stable identification and approximation by signature regression.

Keywords: signatures; log signatures; Lyndon words; signature inversion; term-structure modelling

1 Introduction

Term structures — interest rates, implied volatilities, futures curves — are functions of time-to-maturity that practitioners want to compress into a small number of geometric features. Principal components provide linear summaries of a collection of curves. Signatures instead encode an individual curve through iterated integrals and respect concatenation. For a path $X : [a, b] \rightarrow \mathbb{R}^d$, the signature $S(X)_{a,b}$ collects all iterated integrals along X into a single tensor series; the full series determines X , with its starting point fixed, up to tree-like equivalence (Section 4), and truncated at level m it provides a finite-dimensional encoding of X . The truncated signature contains redundancies coming from the shuffle relations, and the standard remedy is the log signature — whose dimension is that of the free nilpotent Lie algebra $\mathfrak{g}^m(\mathbb{R}^d)$, typically computed in practice through Reizenstein’s `iisignature` library [Reizenstein (2018), Reizenstein and Graham (2020), Reizenstein (2015)].

The question studied here is when a reduced set of signature entries permits identification and stable recovery of a time-augmented path. We first make the coordinate construction explicit: a lexicographic shuffle elimination retains exactly the Lyndon entries, and the remaining entries are polynomial functions of them. Next we separate three inverse problems. Exact recovery asks whether a target belongs to the signature image of a specified path class; we give a complete level-three criterion for two time-ordered segments. Identification asks whether different members of

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that class have different coordinates; polynomial moments and a local rank condition provide two answers. Stability asks how changes in coordinates control changes in the recovered path; on a bounded polynomial family we prove a two-sided estimate with fixed positive weights.

For longer piecewise-linear paths we use weighted nonlinear least squares. This computation motivates, but is not covered by, every stability result above. We distinguish the target-magnitude normalisation used in the experiments from covariance weighting justified by a Brownian-bridge observation model. We then give obstructions to unrestricted all-level weighting and to excessive approximation-rate claims, and prove a topological stability result for a damped all-level metric. The Brazilian DI futures curve and the US Treasury curve illustrate how truncation, representation and the approximation class affect the numerical reconstruction.

The paper is organised as follows. Section 2 recalls signatures, the shuffle product and Chen’s identity. Section 3 defines core signatures and states their identification with Lyndon coordinates, proved in Section S3 of the supplement. Section 4 inverts core signatures into piecewise-linear paths, from the two-segment closed forms to the regression, and develops the moment identity, finite-level stability, covariance weighting and an all-level metric with a continuity guarantee. Section 5 illustrates the inversion on the two curves. The supplementary material submitted with this paper holds the tabulated substitution rules (Section S1), the closed-form two-segment solutions and the three-segment matching system (S2), the proof of Proposition 1 (S3) and the full proofs of the three stability theorems (S4); the text names the section where each is supplied.

The structural identification of core entries with Lyndon coordinates follows from results that are classical in the algebraic-combinatorics literature ([Reutenauer (1993)], the Lyndon basis theorem and Radford’s theorem) and in the rough-paths literature ([Amendola, Friz and Sturmfels (2019)], the universal variety $G^n(\mathbb{R}^d)$). The coordinate theorem is theirs; our contribution is the computational packaging: naming the subset, exhibiting an explicit shuffle-product elimination that produces it, tabulating the substitution rules Q_m , and using these coordinates for Chen-based inversion. For previous signature-inversion algorithms see [Fermanian et al. (2023)].

After our 2023 preprint, [Amendola et al. (2025)] introduced a Macaulay2 package (`PathSignatures`) whose `lyndonShuffle` method implements the same Lyndon-shuffle decomposition that underlies our substitution rules Q_m . In their Section 6 they use the term *core signature tensor* with a different meaning: the signature of a base path that generates a family by linear transformation, suited to the algebraic geometry of signature varieties rather than to Chen-based inversion of individual signatures. The terminology overlap is unfortunate but the constructions are distinct. A parallel Julia/OSCAR implementation is in [Riffo and Schmitz (2026)].

2 Signatures

A path is a continuous map $X : [a, b] \rightarrow \mathbb{R}^d$ of bounded variation; we reparametrise to $[0, 1]$ when convenient. The signature truncated at level m is

$$S^{(\leq m)}(X)_{a,b} = \left(1, S^{i_1}, S^{i_1, i_2}, \dots, S^{i_1, \dots, i_m} \right), \quad i_k \in \{1, \dots, d\}, \quad (1)$$

with level- k entries

$$S^{i_1, \dots, i_k} = \int_a^b \int_a^{t_k} \dots \int_a^{t_2} dX_{t_1}^{i_1} \dots dX_{t_k}^{i_k}. \quad (2)$$

For a single line segment with slopes $(\beta_1, \dots, \beta_d)$ over an interval of length T , every entry has a closed form:

$$S^{i_1, \dots, i_k} = \frac{\beta_{i_1} \dots \beta_{i_k} T^k}{k!}. \quad (3)$$

This is the building block we will use throughout the inversion.

Shuffle product. A (p, q) -shuffle of two ordered sequences of lengths p, q is an interleaving that preserves the relative order within each sequence; $I \sqcup J$ denotes the multiset of all shuffles of multi-indices I and J . The fundamental shuffle identity — valid for any path — is

$$S^I \cdot S^J = \sum_{K \in I \sqcup J} S^K. \quad (4)$$

Equation (4) is the algebraic engine of Section 3: it provides the relations that reduce the truncated signature to its core.

Chen’s identity. For consecutive paths $X : [a, b] \rightarrow \mathbb{R}^d$ and $Y : [b, c] \rightarrow \mathbb{R}^d$, the concatenation $Z = X * Y$ has signature

$$S(Z)_{a,c} = S(X)_{a,b} \otimes S(Y)_{b,c}, \quad \text{or componentwise} \quad z_{i_1, \dots, i_k} = \sum_{j=0}^k x_{i_1, \dots, i_j} \cdot y_{i_{j+1}, \dots, i_k}, \quad (5)$$

with the conventions $x_\emptyset = y_\emptyset = 1$. The componentwise form allows us to compute individual entries of Z without rebuilding the full concatenated signature — the basis of the inversion algorithm in Section 4.

3 Core Signatures

The shuffle relations mean that not every entry of the truncated signature carries new information. A standard parsimonious encoding represents the log signature in a basis of the free Lie algebra, such as a Hall basis. The core construction instead selects a subset of the truncated signature itself — the *core elements* — has the same dimension and the same information content as the log signature, can be read directly from $S(X)$, and admits an explicit structural characterisation in terms of Lyndon words.

3.1 Core variables via the shuffle product

Let $V_m = \{s_w : |w| = m\}$ denote the set of signature entries at level m , and let C_m be the *core set up to level m* , defined by processing words at each level in increasing lexicographic order and retaining an entry when the shuffle identities do not express it as a polynomial in earlier entries at that level and entries at lower levels. Section S3 of the supplement proves that this procedure retains the Lyndon words. For $d = 2$, the recursive application of (4) at level 3 gives the following elimination relations

$$\begin{aligned} & 3s_{1,1,1} - s_1s_{1,1}, \quad -s_2s_{1,1} + s_1s_{2,1} + s_{1,1,2} - s_{2,1,1}, \\ & -s_1s_{2,1} + s_{1,2,1} + 2s_{2,1,1}, \quad -s_2s_{1,2} + s_2s_{2,1} + 2s_{1,2,2} - 2s_{2,2,1}, \\ & -s_2s_{2,1} + s_{2,1,2} + 2s_{2,2,1}, \quad 3s_{2,2,2} - s_2s_{2,2}. \end{aligned}$$

After eliminating six variables in terms of $C_2 = \{s_1, s_2, s_{1,2}\}$ and processing the level-three words in lexicographic order, two entries remain free:

$$C_3 = C_2 \cup \{s_{1,1,2}, s_{1,2,2}\}.$$

Repeating the construction at higher levels yields the core sizes for $d = 2$:

$$|C_m| \text{ for } m = 1, \dots, 6 = 2, 3, 5, 8, 14, 23, \quad (6)$$

matching the log-signature column of Reizenstein’s tables. For $d = 3$ the corresponding counts are 3, 6, 14, 32, 80, 196. The Lyndon multi-indices that constitute C_4 and C_5 are listed in Section S1 of the supplement (and those of C_6 in Table 2), where the 48 substitution rules Q_m for

$m \leq 5$ are tabulated in full. The same elimination procedure extends to higher levels and larger alphabets; the tables stop at level five for $d = 2$.

The companion module `coresig.py` in the repository (see the code-and-data statement) implements the binary word index and the piecewise-linear signature: `Words` identifies Lyndon words, `segment_signature` evaluates a single segment, `chen` concatenates signatures, and `piecewise_signature` computes the full truncated signature of a piecewise-linear path and selects the core. The module uses NumPy and SciPy [Harris et al. (2020), Virtanen et al. (2020)]. The script `listing2_rules_check.py` checks two rules of Section S1 of the supplement on a random four-segment path.

m	$d = 2$ sig	$d = 2$ core/log	$d = 3$ sig	$d = 3$ core/log	$d = 4$ sig	$d = 4$ core/log
1	2	2	3	3	4	4
2	6	3	12	6	20	10
3	14	5	39	14	84	30
4	30	8	120	32	340	90
5	62	14	363	80	1364	294
6	126	23	1092	196	5460	964

Table 1: Sizes of signatures and core signatures at low levels. Core sizes coincide with the log-signature sizes of [Reizenstein (2015)].

3.2 Core elements as Lyndon coordinates

The numerical coincidence $|C_m| = \dim \mathfrak{g}^m(\mathbb{R}^d)$ and the entries retained by lexicographic elimination are consequences of the Lyndon basis of the free Lie algebra. Fix the lexicographic order on $\mathcal{A} = \{1, \dots, d\}$; a non-empty word w is a *Lyndon word* if $w < vu$ for every non-trivial factorisation $w = uv$. For $d = 2$, the Lyndon words of length ≤ 6 on $\{1, 2\}$ are listed in Table 2; comparison with C_6 shows the multi-indices agree exactly.

Length	Lyndon words on $\{1, 2\}$
1	1, 2
2	12
3	112, 122
4	1112, 1122, 1222
5	11112, 11122, 11212, 11222, 12122, 12222
6	111112, 111122, 111212, 111222, 112122, 112212, 112222, 121222, 122222

Table 2: Lyndon words on $\{1, 2\}$ up to length 6. These are exactly the multi-indices of the core variables.

Proposition 1 (Core variables are Lyndon coordinates). *The core sets C_m identified by the shuffle-product computation are exactly the Lyndon words on $\{1, \dots, d\}$ of length $\leq m$. The corresponding σ_w form a transcendence basis for the coordinate ring of the universal variety $G^m(\mathbb{R}^d)$, and the substitution rules Q_m are the explicit form of the corresponding Radford isomorphism truncated at level m .*

The proof uses three classical ingredients: the Lyndon basis theorem for the free Lie algebra ([Reutenauer (1993)] Theorem 5.1), Poincaré–Birkhoff–Witt, and Radford’s theorem that the shuffle algebra is the free commutative algebra on Lyndon words ([Reutenauer (1993)] Theorem 6.1). Together they identify the Lyndon-indexed signature entries with algebraically independent coordinates on the universal variety $G^m(\mathbb{R}^d)$ studied in [Amendola, Friz and Sturmfels (2019)]. The triangular identity associated with Lyndon factorisation explains why lexicographic elimination retains these entries. The proof is given in Section S3 of the supplement.

3.3 Witt formula and the Pascal-triangle structure

The number of Lyndon words of length n on a d -letter alphabet is given by Witt’s formula ([Reutenauer (1993)] Section 4):

$$\ell_d(n) = \frac{1}{n} \sum_{k|n} \mu\left(\frac{n}{k}\right) d^k, \quad (7)$$

where μ is the Möbius function. For $d = 2$ this gives the sequence $2, 1, 2, 3, 6, 9, 18, 30, \dots$ whose cumulative sums match (6). The multi-graded refinement of (7) — counting Lyndon words on $\{1, 2\}$ with prescribed numbers of 1s and 2s — recovers the Pascal-triangle distribution we observed empirically in the substitution rules.

4 Inversion of Core Signatures

We use Chen’s identity (5) to seek a piecewise-linear path with prescribed core coordinates. Exact recovery requires all matching equations to hold and the parameters to define an admissible path. Solving a square subsystem alone need not satisfy the omitted equations. When exact recovery is unavailable, we minimise a weighted residual over a chosen path class. We develop the time-ordered case $X(t) = (t, f(t))$, with $f(0)$ fixed, and also give planar examples.

Related work on the inversion problem. Three recent papers from the Berlin/TU group develop the inversion problem from an algebraic-geometric angle that complements ours. [Amendola, Lotter and Schmitz (2026)] generalise the piecewise-linear and polynomial classes of [Amendola, Friz and Sturmfels (2019)] to piecewise polynomial paths and splines, and introduce the *path recovery degree* (Definition 6.1) as the number of complex points in a generic fibre of the signature map, that is, the number of paths in a class sharing one full truncated signature. For $d = 2$ at level $k = 4$, their Table 3 records the recovery degree as 4 for piecewise linear paths with four segments and 48 for the degree-four polynomial class. [Schmitz (2025)] gives an $O(d^4)$ algorithm for the level-3 d -segment piecewise linear case using Gauss transformations on signature tensors as a tensor analogue of congruence normal forms, outperforming Gröbner-basis methods. [Amendola and Schmitz (2025)] treats the adjacent problem of path recovery from *barycenters* of multiple signatures, relevant when the input is an expected signature.

The time-ordered setting gives explicit recovery formulas for two segments and a polynomial matching system for three (Section S2 of the supplement). It also exposes a moment map inside the signature. This map yields the identification and stability results below and explains which part of signature regression is moment matching. A comparison with a path-space fitting objective is a separate question: neither the moment identity nor identifiability implies equality of the two minimisers.

Remark 2 (What the signature determines). A theorem of [Hambly and Lyons (2010)] identifies the ambiguity of the full signature: for paths of bounded variation with a fixed starting point, equality of signatures is equivalent to *tree-like equivalence*. Immediate out-and-back excursions are a simple example of this equivalence. Each equivalence class contains a tree-reduced path, unique up to reparametrisation, which is the representative of minimal length. Two consequences organise this section. First, a time-ordered path $X(t) = (t, f(t))$ cannot retrace itself, since its first coordinate is strictly increasing, so it is tree-reduced and its full signature determines f once $f(0)$ is fixed (the signature sees only df). Truncating the signature turns this into an inverse problem with its own questions — whether a finite-segment path with the truncated target exists, how many there are (the recovery degrees of [Amendola, Lotter and Schmitz (2026)]), and how stably they depend on the target — which the rest of this section takes up one at a time. Second, the coordinate t in $X(t) = (t, f(t))$ supplies a specified clock, and the entries with a single letter 2 are moments of the increment in that clock (Proposition 5). For a general planar path $(x(s), y(s))$

neither coordinate need be monotone, and the moment interpretation is unavailable. Its signature is invariant under increasing reparametrisation [Chevyrev and Kormilitzin (2016)] and retains the order of traversal. Section 4.3 recovers polygonal approximations to the semicircle and the digit 5 directly from their planar core entries. A clock can also be retained as an additional coordinate, giving $(s, x(s), y(s))$ on the three-letter alphabet; its mixed entries then record the chosen timing. The following proposition records the invariance under an inserted out-and-back excursion.

Proposition 3 (Retracings are invisible). *Let P be a polygonal path in $\mathbb{R}^d$ and let P' be obtained from P by inserting at any knot an excursion along a segment v and straight back, v then $-v$. Then $S(P') = S(P)$ at every level. In particular, for $n \geq 3$ the map from n -segment polygons with fixed endpoints to their signatures is not injective, and for every level m the regression objective (9), as a function of the n -segment polygon, is constant along every d -parameter family $\{P \text{ with } v, -v \text{ inserted} : v \in \mathbb{R}^d\}$; it has flat directions of dimension d at every polygon that contains a retracing. No such family exists for time-ordered paths, whose segments all have positive time increment.*

Proof. The signature of a straight segment with increment v is $\exp(v) = \sum_{k \geq 0} v^{\otimes k} / k!$ (this is (3)), and $\exp(v) \otimes \exp(-v) = 1$ in the tensor algebra. By Chen's identity (5) the excursion contributes the factor $\exp(v) \otimes \exp(-v) = 1$ to the product of segment signatures, so $S(P') = S(P)$. A time-ordered path has $v_1 > 0$ on every segment, so no segment can be followed by its reverse. $\square$

The self-crossing fits of Figure 3 are not exact retracings; Proposition 3 exhibits the ambiguity that motivates the regularisation. Along an exact inserted out-and-back family the signature is constant, while the added length is $2|v|$; a positive penalty on excess length therefore favours $v = 0$ on that family. In a general penalised fit, however, length and signature error are traded off.

4.1 Time-ordered paths: two pieces

Let $X(t) = (t, f(t))$ on $[0, T]$, translated so that $f(0) = 0$. We seek a concatenation $X * Y$ of two line segments

$$X(t)|_{[0, t_1]} = (t, \beta_1 t), \quad Y(t)|_{[t_1, T]} = (t, \beta_1 t_1 + \beta_2 (t - t_1)),$$

parametrised by $\{T, t_1, \beta_1, \beta_2\}$, whose core signature matches a target $\{z_1, z_2, z_{1,2}, z_{1,1,2}, z_{1,2,2}\}$. By (3) and (5), the matching conditions are the five equations of Table 3.

Level	Eq #	Equation
1	1	$T = z_1$
1	2	$\beta_1 t_1 + \beta_2 (T - t_1) = z_2$
2	3	$\frac{1}{2} \beta_1 t_1^2 + \frac{1}{2} \beta_2 (T - t_1)^2 + \beta_2 t_1 (T - t_1) = z_{1,2}$
3	4	$\frac{1}{6} \beta_1 t_1^3 + \frac{1}{6} \beta_2 (T - t_1)^3 + \frac{1}{2} \beta_2 t_1 (T - t_1)^2 + \frac{1}{2} \beta_2 t_1^2 (T - t_1) = z_{1,1,2}$
3	5	$\frac{1}{6} \beta_1^2 t_1^3 + \frac{1}{6} \beta_2^2 (T - t_1)^3 + \frac{1}{2} \beta_2^2 t_1 (T - t_1)^2 + \frac{1}{2} \beta_1 \beta_2 t_1^2 (T - t_1) = z_{1,2,2}$

Table 3: System for the two-piece time-ordered inversion.

The system has five equations in four unknowns. Dropping equation #5 or equation #4 leaves a square system, and each has a closed-form solution (Section S2 of the supplement); we call them Solution 1 and Solution 2. They solve different subsystems, so in general neither matches all five entries: substituting either into its dropped equation leaves a residual. Where

both formulas are defined, those residuals vanish and the two solutions coincide when the target satisfies

$$2z_{1,2}^2 + z_1 z_2 z_{1,2} = 3z_1 z_{1,2,2} + 3z_2 z_{1,1,2}, \quad (8)$$

which is necessary for a two-segment target. Where both formulas are defined they coincide on this hypersurface. The omitted residual is an algebraic consistency diagnostic; it is not a path-space distance. The next statement includes admissibility and the degenerate case.

Proposition 4 (Two-segment recovery). *Let $z = (T, a, b, c, d) = (z_1, z_2, z_{1,2}, z_{1,1,2}, z_{1,2,2})$ with $T > 0$, and put $\Delta = Ta - 2b$. If $\Delta \neq 0$, define*

$$t_* = \frac{2(Tb - 3c)}{\Delta}, \quad R = Tab + 2b^2 - 3Td - 3ac.$$

The first four matching equations have an admissible solution with two positive durations if and only if $0 < t_ < T$. When it exists, this solution is unique, is given by Solution 1 of Section S2 of the supplement, and its residual in equation #5 is $R/(3T)$. Thus z is the core signature of a two-segment time-ordered path if and only if $0 < t_* < T$ and $R = 0$. On this set, Solution 2 gives the same path when $a \neq 0$; its formula is undefined when $a = 0$.*

If $\Delta = 0$, such a path exists if and only if

$$b = \frac{Ta}{2}, \quad c = \frac{T^2 a}{6}, \quad d = \frac{Ta^2}{6}.$$

It is then the straight path $f(t) = at/T$; every interior breakpoint with equal slopes represents that same path.

Proof. For fixed $t_1 \in (0, T)$, equations #2–#3 are linear in the slopes, with determinant $t_1(T - t_1)T/2 \neq 0$. After solving them, equation #4 reduces to $t_1\Delta = 2(Tb - 3c)$. If $\Delta \neq 0$, this forces $t_1 = t_*$. If $t_* \in (0, T)$ the slopes are finite and unique; otherwise no admissible solution exists. Substitution into #5 gives the residual $R/(3T)$. When $a \neq 0$ and $R = 0$, $3Td - 2b^2 = a(Tb - 3c)$, so Solution 2 has the same breakpoint and, by the preceding linear solve, the same slopes.

For any admissible two-segment path, direct expansion gives $\Delta = (\beta_1 - \beta_2)t_1(T - t_1)$. Hence $\Delta = 0$ forces $\beta_1 = \beta_2 = a/T$. The stated coordinates are precisely the straight-segment signature, proving necessity and sufficiency. $\square$

The compatibility equation describes exact image membership only together with admissibility. Its vanishing does not make a boundary value $t_* = 0$ or T an admissible two-segment solution when $\Delta \neq 0$. The case $a = 0$ also illustrates why the two reduced formulas must not be interpreted as two inverse images of one target.

Worked examples. For $f(t) = \sin(\pi t)$ on $[0, 1]$, the target core signature is $\{1, 0, -2/\pi, -1/\pi, 1/4\}$. Solution 1 (dropping equation #5) gives the symmetric tent with peak $(1/2, 4/\pi)$. For $f(t) = \sin(\pi t^2)$ Solution 1 gives an asymmetric tent with its peak at $t_1 = 0.89$, far past the midpoint, matching the asymmetry of the curve (Figure 1a,b); its residual in equation #5 is -0.019 . For $f(t) = \sin(\pi t^2) + 1 - t$ both solutions exist and produce two two-segment fits whose crossing point sits very close to the original curve (Figure 1c). They match different subsets of coordinates, as the preceding proposition explains.

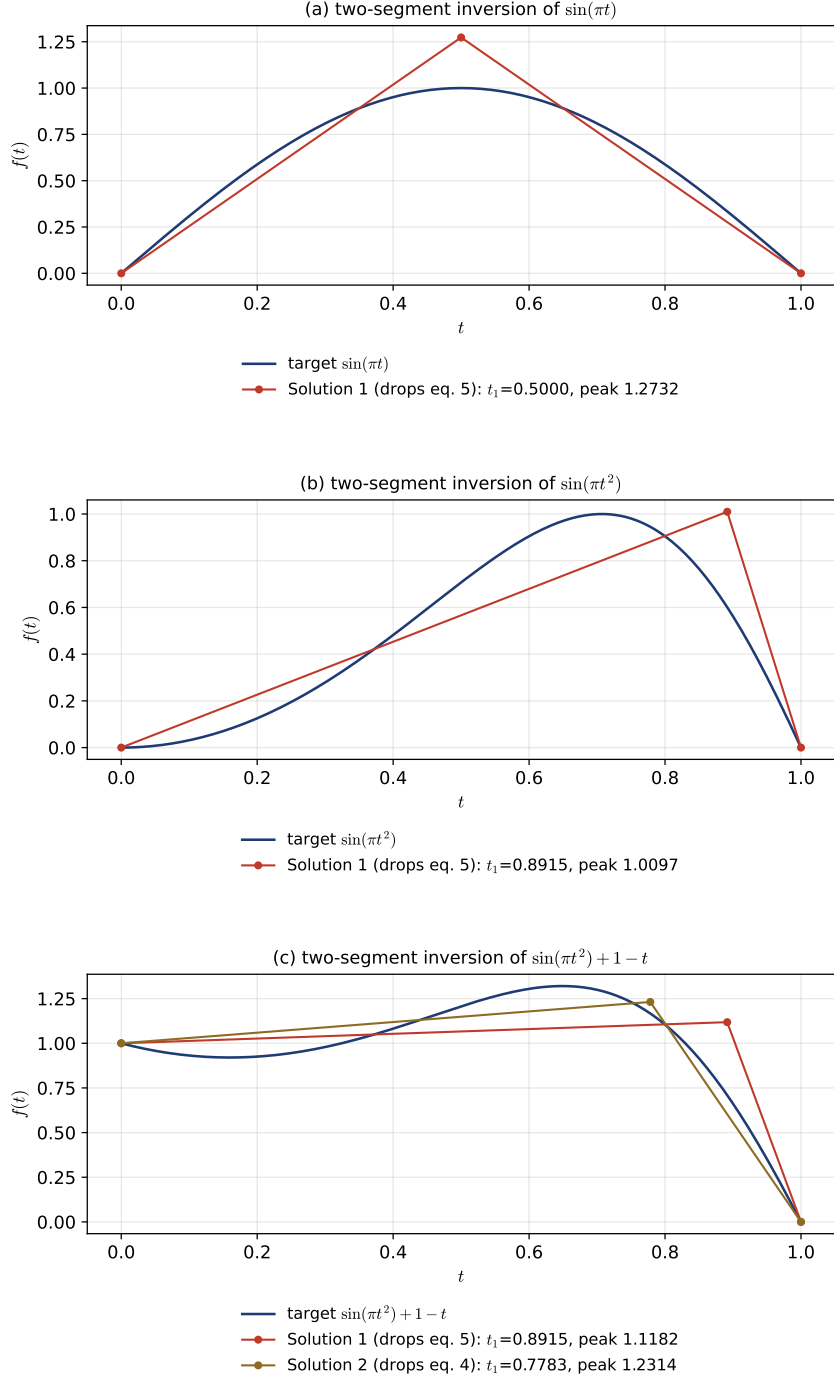


Figure 1: Two-segment inversions (`two_piece_inversion` in `coresig.py`). (a) $\sin(\pi t)$: Solution 1 is the symmetric tent with peak $(1/2, 4/\pi)$; Solution 2 does not exist because $z_2 = 0$. (b) $\sin(\pi t^2)$: Solution 1 places the peak at $t_1 = 0.8915$. (c) $\sin(\pi t^2) + 1 - t$: both solutions exist, with $t_1 = 0.8915$ and 0.7783 , and cross close to the curve.

The routine `two_piece_inversion` in `coresig.py` inverts a two-segment target numerically instead of through the closed forms of Section S2 of the supplement: the duration is $T = z_1$, equations #2–#3 are linear in the two slopes for a given breakpoint, and the remaining kept equation is solved for the breakpoint by bracketing; every root is returned with the residual of the dropped equation, which tests compatibility with the omitted coordinate.

4.2 Adversarial cases and three pieces

A vanishing Δ does not by itself imply failure: it gives exact recovery when the target is a straight line. For an oscillatory target such as $f(t) = \sin(2\pi t)$, however, the level-two coordinates agree with the chord while higher coordinates do not. Proposition 4 then excludes a two-segment representation. A third segment enlarges the candidate class without guaranteeing exact recovery. At level four the matching system has eight equations in six unknowns, including T (Section S2 of the supplement). Dropping two equations leaves a square system but does not guarantee an admissible solution or satisfaction of the omitted equations. From three segments onward we use the regression below with all retained coordinates.

4.3 Regression over piecewise-linear paths

For four or more segments the algebraic Gröbner approach becomes prohibitively expensive. The practical alternative is a nonlinear weighted least-squares regression in signature space: with θ parameterising the piecewise-linear path and $\mathcal{L}_m$ the set of Lyndon words up to level m ,

$$\theta^* = \arg \min_{\theta} \sum_{w \in \mathcal{L}_m} w_w (\sigma_w[g\theta] - \sigma_w[X])^2. \quad (9)$$

The experiments use level-only weights $w_w = W_{|w|}$. Table 4 compares the hand-chosen schedule $\{50, 50, 30, 10, 10, 3, 3, 3\}$, target-magnitude normalisation and uniform weights. The magnitude schedule is used in the other numerical illustrations. For a path of total variation V , the factorial bound $|\sigma_w| \leq V^{|w|}/|w|!$ motivates attention to scaling, but does not determine statistically optimal weights.

For a theoretical constrained problem, one may require durations $\Delta t_i \geq \delta > 0$, $n\delta \leq T$, and slopes $|\beta_i| \leq B$. This gives a compact parameter set. The signature map and its weighted residual are continuous, so a minimiser exists on this set; neither uniqueness nor successful recovery by a local solver follows from that argument.

The routine `weighted_regression` in `coresig.py` solves a related penalised problem. Its $2n - 1$ parameters are the interior breakpoints and slopes; a finite penalty discourages durations below `dt_min`, and the slopes and individual knots have bounds; the penalty is finite, so short durations are discouraged, not excluded. SciPy’s bounded trust-region reflective solver is run from several initial points; the routine reports the starts that meet its stopping criteria and returns the lowest-cost result found, and meeting those criteria does not certify a local minimum. The function name `variance_equalising_weights` is retained in the code for compatibility with earlier runs; it implements the magnitude schedule defined in Section 4.5.

Figure 2 shows four- and five-segment fits to two oscillatory time-ordered paths; Figure 3 takes the regression to two paths that are not time-ordered.

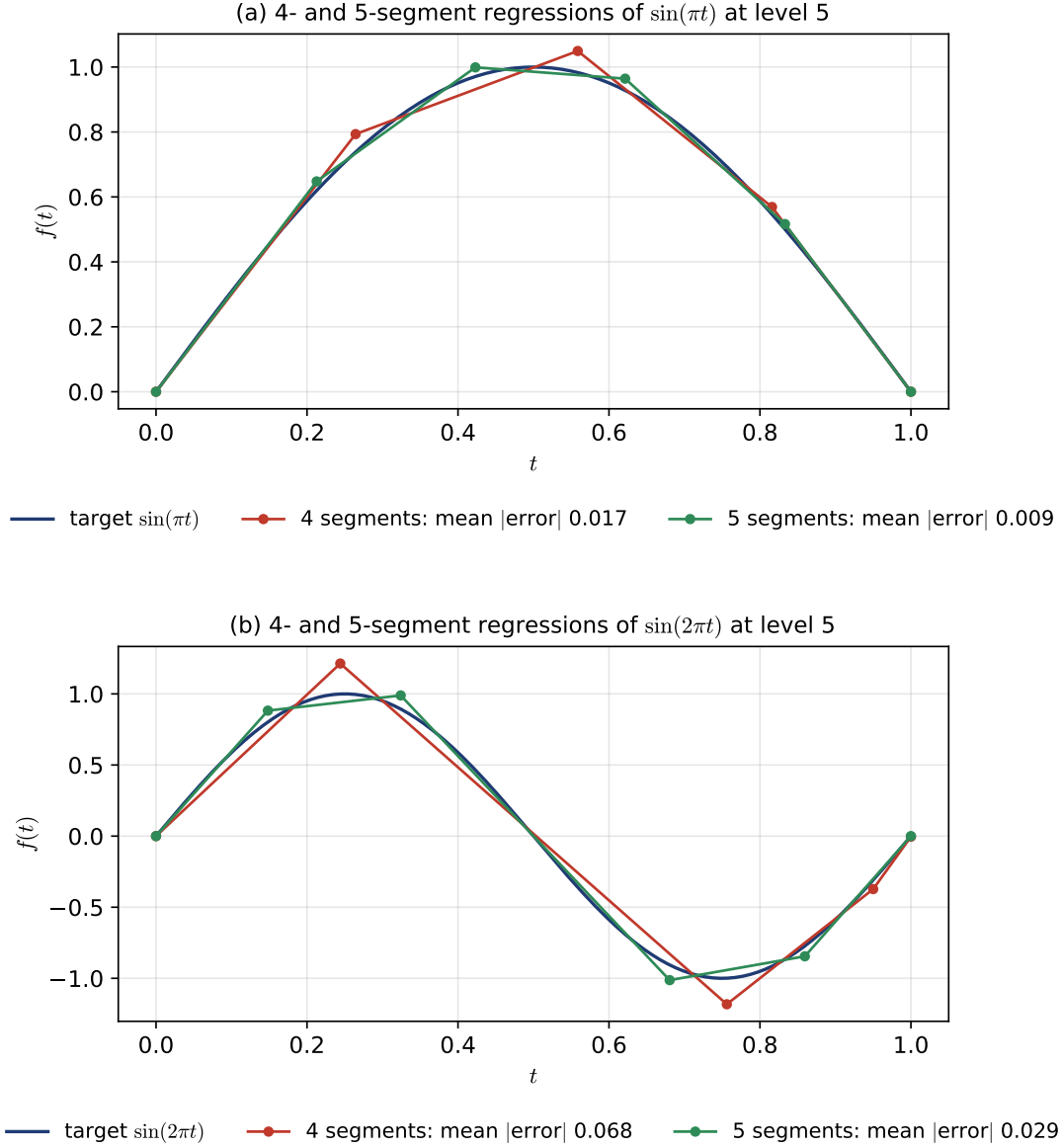


Figure 2: Four- and five-segment regressions (`weighted_regression`) at level 5 with magnitude-normalised weights, 24 starts. (a) $\sin(\pi t)$: mean absolute error 0.017 with four segments, 0.009 with five. (b) $\sin(2\pi t)$: 0.068 and 0.029; one full period asks for more than four segments, and the four-segment fit puts a breakpoint at the minimum-duration penalty threshold.

General two-dimensional paths, with the knots free in both coordinates, use the same regression with $2(n - 1)$ unknowns; with the starting point at the origin, the level-1 entries fix the endpoint. The routine `weighted_regression_2d` matches the planar signature of the ordered vertices, and segment durations do not enter its objective. Without a clock coordinate the entries are no longer moments, and the semicircle and the digit 5 illustrate inversion in this planar formulation. On the semicircle, the five-segment regression at level 5 finds a self-crossing polygon, 1.96 chords long, with a near-retracing at the top of the arc, whose core entries match the arc's to a residual of 5×10^{-7} , three hundred times smaller than the near-arc solution's; exact out-and-back excursions cancel at every level (Proposition 3), and the near-retracing in this fit comes with a small finite-level residual. Adding one residual, a small multiple of the path length in excess of the chord (coefficient 0.03 in the units of the weighted residuals), penalises longer polygons and yields a fit following the arc in this experiment (Figure 3): five segments within 0.047 of the arc, length 1.60 chords against the arc's 1.57. The digit-5 fit uses the same planar

objective (Figure 3b). A clock-dependent inversion retains a chosen clock as an additional coordinate — arc length, interpolated vertex index, or a spatial coordinate that is strictly monotone along the traversal — and its mixed entries can change the fitted polygon; the present planar fits use spatial entries only. The next results concern identification and stability for paths with a specified time coordinate.

4.4 Signature entries as iterated path moments

For $X(t) = (t, f(t))$ on $[0, T]$ with f absolutely continuous, write $M_k(f) = \int_0^T t^k f(t) dt$ for the moments of f and $F(t) = \int_0^t f$. The core entries of X split into two families: the words $(1^{k-1}, 2)$ with a single letter 2, and the words with two or more.

Proposition 5 (Single-letter-2 entries are the moments of df). *For every $k \geq 1$,*

$$\sigma_{(1^{k-1}, 2)}[X] = \frac{1}{(k-1)!} \int_0^T t^{k-1} df(t) = \frac{T^{k-1}}{(k-1)!} f(T) - \frac{M_{k-2}(f)}{(k-2)!} \quad (k \geq 2).$$

In particular $\sigma_2[X] = f(T) - f(0)$, $\sigma_{1,2}[X] = Tf(T) - M_0(f)$ and $\sigma_{1,1,2}[X] = \frac{T^2}{2}f(T) - M_1(f)$.

Proof. The iterated integral over $0 < t_1 < \dots < t_{k-1} < t_k < T$ of $dt_1 \dots dt_{k-1} df(t_k)$ integrates the first $k-1$ variables to $t_k^{k-1}/(k-1)!$. Integration by parts against df gives the boundary term (the $t=0$ term vanishes for $k \geq 2$) minus $\int_0^T t^{k-2} f dt/(k-2)!$. $\square$

Corollary 6 (D levels recover a degree- D curve). *Let $f(t) = \sum_{i=1}^D \alpha_i t^i$ on $[0, 1]$. The D entries $\sigma_{(1^{l-1}, 2)}[X]$, $l = 1, \dots, D$, determine α by one linear solve: $\sigma_{(1^{l-1}, 2)} = \frac{1}{(l-1)!} \sum_i \alpha_i \frac{i}{l+i-1}$, and the matrix $(i/(l+i-1))_{l,i}$ is nonsingular.*

Proof. $\int_0^1 t^{l-1} d(t^i) = i \int_0^1 t^{l+i-2} dt = i/(l+i-1)$. The matrix is the Hilbert matrix $(1/(l+i-1))$ with column i scaled by i ; the Hilbert matrix is nonsingular (it is a Cauchy matrix), and column scaling preserves that. $\square$

This is recovery within a specified polynomial family; a parameter count alone gives no corresponding theorem for another family. Conditioning also matters. At $D = 5$, the column-scaled Hilbert matrix in the corollary has condition number about 2.8×10^5 , compared with 4.8×10^5 for the Hilbert matrix. The unscaled signature map includes factorial row factors and can be worse conditioned. Additional core entries provide additional residuals; whether they improve conditioning or lower the recovery depth is a property of the family, treated in Theorem 9.

On a polynomial family the moment map also gives a stability statement at a fixed level, which is separate from the infinite-dimensional question at the end of Section 4.5.

Theorem 7 (Finite-level stability on polynomial families). *Fix $T > 0$, an integer $D \geq 1$, an exponent $s \geq 0$ and positive weights q_w for the Lyndon words w of length at most D . Let $\mathcal{P}_D^0$ be the polynomials of degree at most D with $f(0) = 0$, let $K \subset \mathcal{P}_D^0$ be bounded, and for $f, g \in \mathcal{P}_D^0$ put*

$$E_D(f, g) = \sum_{|w| \leq D, w \in \mathcal{L}} q_w (\sigma_w[X_f] - \sigma_w[X_g])^2, \quad X_f = (t, f(t)).$$

Then there are constants $0 < c \leq C < \infty$, depending on D, T, s , the weights and (for C) the size of K , such that

$$c \|f - g\|_{H^s(0, T)}^2 \leq E_D(f, g) \leq C \|f - g\|_{H^s(0, T)}^2 \quad \text{for all } f, g \in K.$$

The constant c can be chosen independently of K .

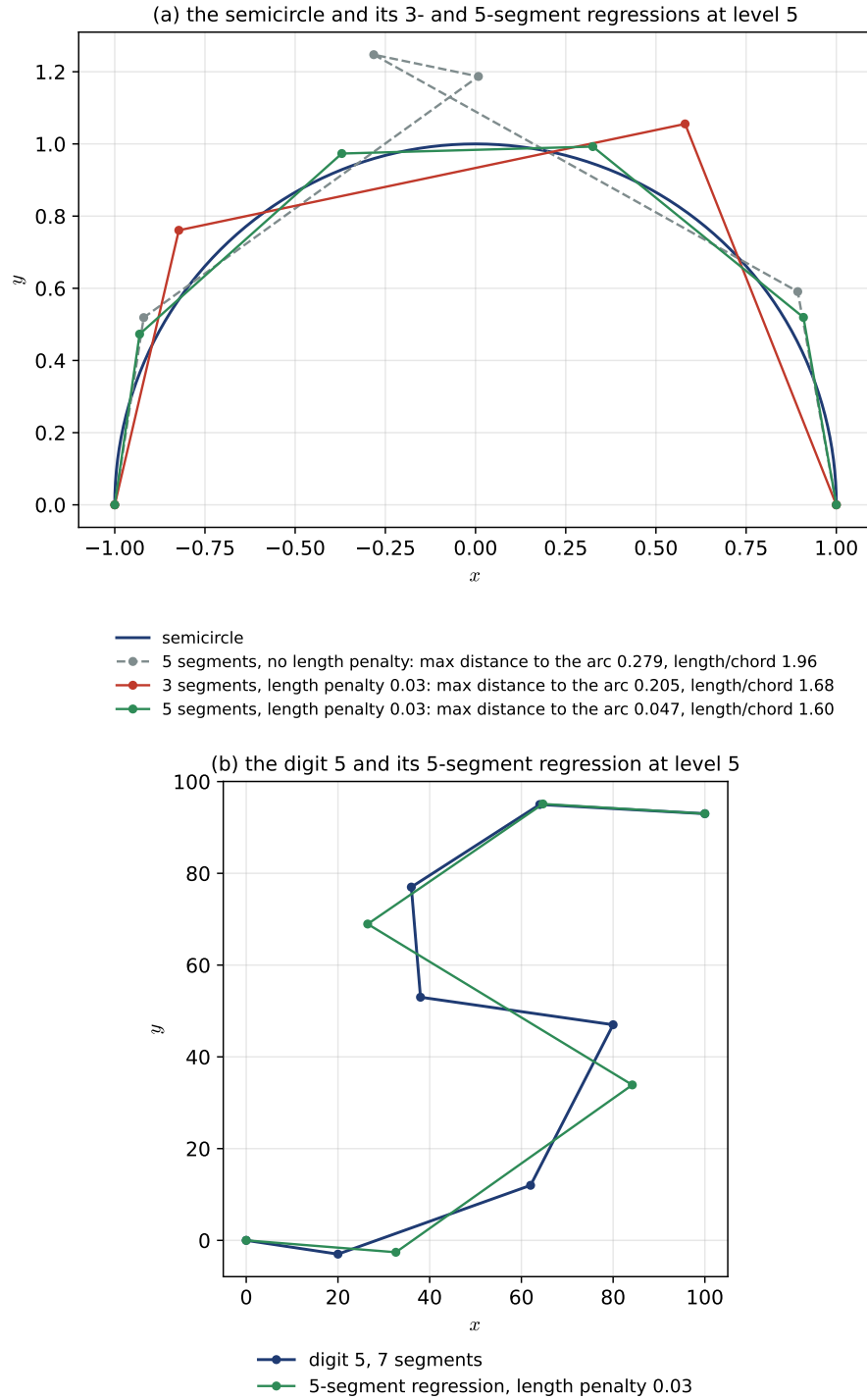


Figure 3: General two-dimensional paths. (a) The semicircle and its regressions at level 5, magnitude-normalised weights, 24 starts: five segments without the length penalty (grey, dashed) is the self-crossing polygon with the smallest signature residual, 1.96 chords long, with a near-retracing at the top; three and five segments with the penalty follow the arc, the five-segment one within 0.047 and 1.60 chords long against the arc’s 1.57. (b) A seven-segment path tracing the digit 5 (knots in the repository file `digit5_points.csv`) and its five-segment regression with the penalty, which recovers the topology of the stroke.

Sketch. The D single-letter-2 coordinates are $M\alpha$ with M the scaled Hilbert matrix of Corollary 6, invertible, which gives a positive lower bound in coefficient norm, $E_D(f, g) \geq q_{\min} s_{\min}(M)^2 \|\alpha - \alpha'\|^2$; every other retained coordinate is a polynomial in α , so the mean-value inequality on a ball containing K gives an upper bound in the same norm; and on the finite-dimensional space $\mathcal{P}_D^0$ the coefficient norm and the H^s norm are equivalent, which converts both bounds into the stated ones (the constants c, C below include that conversion). The full proof is in Section S4 of the supplement. $\square$

The theorem is a finite-dimensional consequence of the moment map; its value is the statement with its hypotheses, not the mechanism. The displayed lower bound can deteriorate with D through $s_{\min}(M)$, which is 3.9×10^{-3} at $D = 3$ and 1.2×10^{-6} at $D = 5$ for $T = 1$ (the $1/(l-1)!$ row scaling makes M worse conditioned than the matrix of Corollary 6), and target-dependent weights such as (10) are covered only where they admit uniform positive lower and finite upper bounds on K .

The constants can be specified in coefficient coordinates. Let H_s be the Gram matrix of $t, t^2, \dots, t^D$ for the chosen $H^s(0, T)$ inner product, let Q contain the weights of the D moment coordinates, and let $\Psi(\alpha)$ be the full vector of weighted core entries $\sqrt{q_w} \sigma_w[X_f]$. On a convex closed coefficient ball B_R containing K , valid choices are

$$c = \lambda_{\min} \left(H_s^{-1/2} M^\top Q M H_s^{-1/2} \right), \quad C = \sup_{\alpha \in B_R} \left\| D\Psi(\alpha) H_s^{-1/2} \right\|_{\text{op}}^2.$$

The first follows from the moment subvector; the second follows by integrating the derivative along a coefficient line segment. These formulas expose dependence on the norm, degree, weights and parameter range. They are not uniform-in-degree estimates.

Corollary 8 (What stability gives a regression). *Under the hypotheses of Theorem 7, let $f \in K$ and let $A \subset K$ be a nonempty candidate class. If $\hat{g} \in A$ satisfies $E_D(f, \hat{g}) \leq \inf_{g \in A} E_D(f, g) + \varepsilon$, then*

$$\|f - \hat{g}\|_{H^s}^2 \leq \frac{C}{c} \inf_{g \in A} \|f - g\|_{H^s}^2 + \frac{\varepsilon}{c}.$$

Proof. $c\|f - \hat{g}\|^2 \leq E_D(f, \hat{g}) \leq E_D(f, g) + \varepsilon \leq C\|f - g\|^2 + \varepsilon$ for every $g \in A$; take the infimum over g . $\square$

The bound separates the representation error of the class, the stability ratio C/c and the optimisation gap ε , and says nothing about the minimisers coinciding. It applies inside one polynomial family; it does not cover the piecewise-linear fits of Sections 4.3 and 5, and a multi-start search does not certify a gap ε .

Theorem 9 (Local identification and stability). *Let $\theta \mapsto f_\theta$ be an anchored family indexed by an open set $\Theta \subset \mathbb{R}^p$. Suppose every core coordinate is C^1 in θ , and the vector $S_m(\theta)$ of coordinates through level m has Jacobian of rank p at θ_0 . Then some p of these coordinates define a C^1 diffeomorphism from a neighbourhood U of θ_0 onto an open set. Every other core coordinate is a C^1 function of the selected coordinates on U . After reducing U , there are $a, b > 0$ such that*

$$a|\theta - \eta| \leq |S_m(\theta) - S_m(\eta)| \leq b|\theta - \eta| \quad (\theta, \eta \in U).$$

If, additionally, $\theta \mapsto f_\theta$ is C^1 into a chosen $H^s(0, T)$ space with injective derivative at θ_0 , the parameter distance in this estimate can be replaced by $\|f_\theta - f_\eta\|_{H^s}$, with different constants.

Sketch. Apply the inverse function theorem to an invertible p -row minor F of $DS_m(\theta_0)$; every other coordinate factors through F^{-1} . On a small convex ball the variation of DF is below half the smallest singular value of $DF(\theta_0)$, and integrating derivatives along segments gives the two-sided estimate; the H^s version uses the injectivity of the path derivative. The full proof is in Section S4 of the supplement. $\square$

For a fixed parameterisation, define its local identification depth at a point as the smallest m satisfying this rank condition, when such an m exists. Redundant parameters must first be removed. The theorem is local and conditional: it does not supply a rank condition from a parameter count, or prove global identifiability. Higher coordinates are locally redundant for exact identification but may remain useful for estimation with noise.

Remark 10 (Entries with more than one letter 2; the depth of a curve). Lyndon words with two or more letter 2s are iterated integrals of f against itself, and are not determined by the finitely many moments retained at a given level (the full moment sequence determines f on a compact interval, and with it every entry); for instance

$$\sigma_{1,2,2}[X] = \int_0^T \left(\int_0^u t df(t) \right) df(u) = \frac{T}{2} f(T)^2 + \frac{1}{2} \int_0^T f^2 - f(T) F(T),$$

which depends on $\int f^2$. On a fixed finite-dimensional family, such coordinates can contribute to the rank of the signature map. Whether they reduce identification depth must be checked for that family. A Nelson–Siegel–Svensson curve has an additive level parameter invisible to the signature. After fixing that level, five shape parameters remain; the numerical rank-five moment Jacobians found at three parameter sets are local evidence at those points; generic rank, and the behaviour at degenerate decay parameters or vanishing loadings, remain to be established. Theorem 9 states what a verified rank condition gives. The market interpolants in Section 5 belong to no named family; their fit-error sweep tests a numerical choice of truncation.

Recovering a function from finitely many moments is an inverse problem with substantial conditioning difficulties [Talenti (1987)]. The first m moments of f determine its L^2 projection onto polynomials of degree below m . For an integer $r \geq 1$ and $f \in C^r([0, T])$, polynomial approximation estimates imply an $O(m^{-r})$ error for that projection in L^2 [DeVore and Lorentz (1993), Chapter 7]. This is an approximation result for polynomials of increasing degree, not a rate for a fixed-segment signature regression. Matching increment moments also fixes ordinary moments only after the relevant endpoint increment is matched and the initial value is fixed.

Path-space evaluation. A signature residual and a path-space error measure different quantities. For time-ordered paths, a direct numerical benchmark is to minimise

$$\frac{1}{N+1} \sum_{i=0}^N |f(t_i) - g_\theta(t_i)|, \quad t_i = iT/N,$$

over the same candidate class and constraints. This is a sampled mean absolute error, approximating $T^{-1} \|f - g_\theta\|_{L^1}$ as the grid is refined. A numerical search supplies a benchmark; the best value it finds is an upper bound on the minimum. Corollary 8 provides an approximation-quality comparison only on the polynomial classes covered by its hypotheses. No equality of minimisers is claimed for the piecewise-linear experiments.

4.5 Weights, and what the regression minimises

Target-magnitude normalisation. For each level, let a_l be the largest absolute target *core* coordinate at that level. The implemented schedule is

$$a_l = \max_{\substack{w \in \mathcal{L}_m \\ |w|=l}} |\sigma_w[X]|, \quad A = \max_{w \in \mathcal{L}_m} (|w|! |\sigma_w[X]|)^{1/|w|}, \quad \widetilde{W}_l = \frac{1}{\max(a_l, 0.1A^l/l!)^2}, \quad W_l = \frac{\widetilde{W}_l}{\widetilde{W}_1}. \quad (10)$$

The floor prevents a vanishing level from receiving an infinite weight. The time coordinate ensures $A > 0$ in the time-ordered experiments. This schedule scales residuals relative to target

magnitudes; it is not a derived residual covariance. It also need not equalise contributions from different levels, which have different numbers of core coordinates. Changing m can change A and hence weights at earlier levels where the floor is active. The working and uniform schedules in Table 4 provide comparators. None of these numerical comparisons establishes an optimal schedule for a general noise model.

For contrast, the following calculation gives inverse marginal variances under a specific stochastic model.

Proposition 11 (Brownian-bridge moment variances). *Let B be a standard Brownian bridge on $[0, 1]$ and $X = (t, B)$. For $l \geq 2$,*

$$\text{Var } \sigma_{(1^{l-1}, 2)}[X] = \frac{1}{((l-2)!)^2 l^2 (2l-1)},$$

$$W_l^{\text{bridge}} := \frac{1}{\text{Var } \sigma_{(1^{l-1}, 2)}} = 12, 45, 448, 8100, \dots \sim \frac{2(l!)^2}{l}.$$

Proof. The bridge has unbounded variation; the single-letter-2 entries $\frac{1}{(l-1)!} \int_0^1 t^{l-1} dB_t$ have deterministic integrands and are Wiener integrals, for which integration by parts holds as for Proposition 5. With $B_0 = B_1 = 0$, $\sigma_{(1^{l-1}, 2)} = -\frac{1}{(l-2)!} \int_0^1 t^{l-2} B_t dt$. With $a = l-2$ and $\text{Cov}(B_s, B_t) = \min(s, t) - st$, the double integral of $s^a t^a (\min(s, t) - st)$ over $[0, 1]^2$ is $\frac{2}{(a+2)(2a+3)} - \frac{1}{(a+2)^2} = \frac{1}{(a+2)^2(2a+3)}$, and $(a+2)^2(2a+3) = l^2(2l-1)$. Stirling's formula gives the asymptotic. $\square$

The inverse marginal variances grow factorially, with the additional factor $2/l$ shown above. This calculation concerns one coordinate per level and does not identify the covariance of all core residuals. Nor are these moment coordinates independent, as the next result shows. The distinction matters when interpreting weighted least squares [Aitken (1935)].

Proposition 12 (Bridge covariance of the moment coordinates). *With B and $Z_l := \sigma_{(1^{l-1}, 2)}[(t, B)]$ as in Proposition 11, for $r, s \geq 2$,*

$$\text{Cov}(Z_r, Z_s) = \frac{1}{(r-1)!(s-1)!} \left(\frac{1}{r+s-1} - \frac{1}{rs} \right).$$

For every $m \geq 2$ the covariance matrix of $(Z_2, \dots, Z_m)$ is positive definite. If a curve is observed as $f + \varepsilon B$, the perturbations of its moment coordinates are exactly εZ_l , so their covariance is ε^2 times this matrix, and the covariance-weighted residual on the moment coordinates is the generalised least-squares objective for that family under that model.

Proof. Write $B_t = W_t - tW_1$ with W a Brownian motion. Then $Z_l = \frac{1}{(l-1)!} \int_0^1 (t^{l-1} - \frac{1}{l}) dW_t$, since $\int_0^1 t^{l-1} d(tW_1) = W_1/l$, and the Itô isometry gives $\text{Cov}(Z_r, Z_s) = \frac{1}{(r-1)!(s-1)!} \int_0^1 (t^{r-1} - \frac{1}{r})(t^{s-1} - \frac{1}{s}) dt$, which is the formula; the diagonal is Proposition 11. The matrix is the Gram matrix of the functions $t^{l-1} - 1/l$, $l = 2, \dots, m$, in $L^2(0, 1)$, scaled by nonzero factorials; these functions are linearly independent (distinct degrees), so the Gram matrix is positive definite. The last statement is the linearity of $\sigma_{(1^{l-1}, 2)}$ in df (Proposition 5). $\square$

The off-diagonal entries are large: $\text{Corr}(Z_2, Z_3) = \sqrt{15}/4$. So under the bridge model the justified objective on the moment coordinates is the quadratic form of the inverse covariance, and the diagonal weights of Proposition 11 are its marginals only. The proposition says nothing about the entries with more than one letter 2, which are nonlinear in the path, nor about the level-1 entry, which the bridge leaves unchanged and which must be treated separately. A covariance calculation for the nonlinear coordinates requires an integration convention and further analysis; the schedule (10) used in the experiments remains a magnitude normalisation.

Does the schedule matter? Table 4 reports errors that the signature objective does not directly minimise. For the six smooth-function configurations, the mean absolute errors differ by less than 5% across schedules; maximum errors differ by up to about 10%. This does not imply that the fitted polygons or their breakpoints coincide. On DI1 the magnitude schedule reduces the maximum error from 14.8 bp under the working weights to 6.5 bp; uniform weights give 6.9 bp. On the Treasury curve the magnitude schedule gives 10.4 bp, compared with 9.2 and 9.3 bp for the other schedules.

Termination counts also depend on the schedule. Magnitude normalisation has the largest counts on the two curves and on $\sin(2\pi t)$, while it has the smallest counts on $\sin(\pi t^2)$. These are outcomes of a twelve-start experiment. Weighting changes the objective as well as its numerical scaling.

Target	n	mean error			max error			starts converged (of 12)		
		work.	magn.	unif.	work.	magn.	unif.	work.	magn.	unif.
$\sin(\pi t)$	4	0.0161	0.0168	0.0161	0.0657	0.0647	0.0659	5	5	4
$\sin(\pi t)$	5	0.0093	0.0093	0.0092	0.0340	0.0351	0.0339	3	6	3
$\sin(2\pi t)$	4	0.0670	0.0694	0.0670	0.2302	0.2095	0.2259	6	10	4
$\sin(2\pi t)$	5	0.0292	0.0292	0.0292	0.1032	0.1059	0.1036	4	8	7
$\sin(\pi t^2)$	4	0.0179	0.0181	0.0178	0.0740	0.0708	0.0730	9	7	10
$\sin(\pi t^2)$	5	0.0123	0.0126	0.0123	0.0423	0.0403	0.0425	5	4	6
DI1 (bp)	5	3.9	1.9	2.2	14.8	6.5	6.9	2	6	2
H.15 (bp)	4	3.9	4.2	3.9	9.2	10.4	9.3	4	11	5

Table 4: Three weight schedules, level 5, twelve starts, segment-duration penalty threshold 2%: the working values $\{50, 50, 30, 10, 10, 3, 3, 3\}$, the magnitude weights (10), and uniform weights. Pointwise error on a 401-point grid for the functions (units of the target), at the market points for the curves (bp); breakpoints and costs in the repository file `weights_comparison.csv`.

An all-level metric and its limits. Finite-level norm equivalence on a specified polynomial space does not pass automatically to an infinite-dimensional class.

Proposition 13 (An obstruction to undamped weighting). *Let $W_k > 0$ satisfy $\sum_{k \geq 2} W_k/(k!)^2 = \infty$. Then the weighted squared core difference over all levels diverges between $X = (t, 0)$ and either $G = (t, at)$ or $G = (t, at(1 - t))$ on $[0, 1]$, for $a \neq 0$. The second pair has identical endpoints.*

Proof. The single-letter-2 differences are $a/k!$ for $g(t) = at$ and $a(1 - k)/((k + 1)k!)$ for $g(t) = at(1 - t)$. Summing their weighted squares proves the first assertion directly. In the second case the factor $(k - 1)^2/(k + 1)^2$ tends to one, giving the same conclusion. $\square$

This covers weights proportional to $(k!)^2$ and also the inverse bridge variances, for which $W_k/(k!)^2 \sim 2/k$. The obstruction is finiteness, before any question of a Sobolev lower bound. Damping the weights gives a well-defined distance with a qualitative recovery guarantee.

Theorem 14 (A damped core metric on a Lipschitz class). *Fix $T > 0$, $L > 0$, set $A = \max(1, L)$, and let*

$$K_L = \{f \in C([0, T]) : f(0) = 0, |f(t) - f(u)| \leq L|t - u|\}.$$

For $0 < \rho < 1/(\sqrt{2}A)$, define

$$d_\rho(f, g)^2 = \sum_{w \in \mathcal{L}} \frac{(|w|!)^2 \rho^{2|w|}}{T^{2|w|}} |\sigma_w[(t, f)] - \sigma_w[(t, g)]|^2.$$

This series converges uniformly on $K_L \times K_L$. Its square root is a metric on K_L and induces the uniform topology. In particular, there is a nondecreasing function ω with $\omega(r) \rightarrow 0$ as $r \downarrow 0$ such that

$$\|f - g\|_\infty \leq \omega(d_\rho(f, g)) \quad (f, g \in K_L).$$

Sketch. Simplex integration gives $|\sigma_w| \leq A^k T^k / k!$ for a word of length k , so the level- k weighted squared differences are at most $4(2A^2 \rho^2)^k$ and the series converges; d_ρ is an ℓ^2 norm of a difference, hence a metric, and $d_\rho(f, g) = 0$ forces all moments of $f - g$ to vanish (Proposition 5), so $f = g$. Each iterated integral is continuous for uniform convergence under a common variation bound, by induction on the word length; K_L is compact by Arzelà–Ascoli, so the continuous injection is a homeomorphism onto its image and its inverse is uniformly continuous. The full proof is in Section S4 of the supplement. $\square$

The theorem establishes topological recovery on a fixed compact class; a Lipschitz or Hölder inverse, a Sobolev norm equivalence, and a bound uniform in L are the stronger statements that remain open on an infinite-dimensional class.

Open questions. For the metric above, determine quantitative bounds on the inverse modulus ω , and identify additional assumptions under which an inverse estimate in a specified path norm holds. For finite-level piecewise-linear recovery, determine the parameter domains and levels on which a full-rank Jacobian gives useful stability constants, accounting for coincident slopes and collapsing durations. Similar numerical fits from two regression objectives are evidence for none of these.

Approximation barrier. The rate at which a piecewise-linear approximant converges does not improve with the level at which the signature is read:

Proposition 15 (Approximation barrier). *For $f(t) = t^2$ on $[0, 1]$ and any piecewise-linear g with N pieces, continuous or not, $\int_0^1 |f - g| dt \geq 1/(16N^2)$.*

Proof. On an interval of length h centred at c , write $t = c + x$; then $t^2 - (\text{affine in } t) = x^2 - (\text{affine in } x)$, and the best L^1 affine approximation of x^2 on $[-h/2, h/2]$ has slope zero by symmetry (the objective is convex in the affine coefficients and invariant under $x \mapsto -x$) and intercept equal to the median of x^2 under the uniform law, $h^2/16$; its error is $\int_{-h/2}^{h/2} |x^2 - h^2/16| dx = h^3/16$. Summing over pieces of lengths h_i with $\sum h_i = 1$ and using $\sum h_i^3 \geq N(1/N)^3$ gives the bound. $\square$

Thus no rate faster than N^{-2} can hold uniformly for this piecewise-linear approximation problem on a smooth target class containing t^2 (individual targets, such as affine functions, are recovered exactly). A faster rate needs pieces of higher degree, a different model class.

4.6 What the estimates establish

The results distinguish four conclusions. Lyndon coordinates remove algebraic redundancy without discarding truncated-signature information. Exact recovery additionally requires membership of a specified path family’s signature image. Finite-level stability holds on the polynomial families of Theorem 7 and locally under the hypotheses of Theorem 9. The all-level metric of Theorem 14 gives a qualitative inverse modulus on a Lipschitz class. None of these statements identifies weighted signature regression with a Sobolev orthogonal projection, or certifies the numerical optimiser used in the examples.

5 Two term structures: an illustration

This section runs the inversion of Section 4 on one date each of two term structures, illustrating the approximation problem and the dependence of the fit on truncation and representation: the Brazilian one-day interbank deposit futures curve (DI1, traded and settled at B3; settlement rates from [B3 (2023)]) and the US Treasury constant-maturity curve published in the Federal

Reserve’s H.15 release [Federal Reserve (2023)]. The DI1 example contains 37 listed contracts extending to about fourteen years. The Treasury example uses eleven constant-maturity tenors and a public data source. The dates are 7 November 2023 for DI1 and 11 October 2023 for H.15, the last dates in the two data files. The code is the Python module of Section 3 and the scripts in the repository.

Representation. A curve on a given date is the time-augmented path $X(u) = (u, x(u))$, where x is the rate in percent relative to its first value (the level is stored separately; the signature sees only dx) and u is the maturity axis mapped to $[0, 1]$ by $u = (\sqrt{\tau} - \sqrt{\tau_0})/(\sqrt{\tau_{\max}} - \sqrt{\tau_0})$. The square root spreads the short end, where most of the shape lives; its effect on the regression is discussed in Section 5.3. The target signature is that of the linear interpolant of the market points. The weights are the magnitude-normalisation schedule (10) computed from that signature. The regression is the multi-start weighted least squares of Section 4.3, with 24 random starts and a penalty threshold of 2% of the path for segment duration. The level m is the smallest at which the system is overdetermined, $|C_m| - 1 > 2n - 1$; the number of segments n is chosen from the sweep of Section 5.3, which compares n and $n - 1$ for each curve and stops at the count that reduces the reported error substantially relative to the smaller count. Larger counts were not run; the choices are illustrative. The script `listing5_curve.py` in the repository shows the inversion call for one date, using the companion module and data loader.

5.1 The DI1 curve

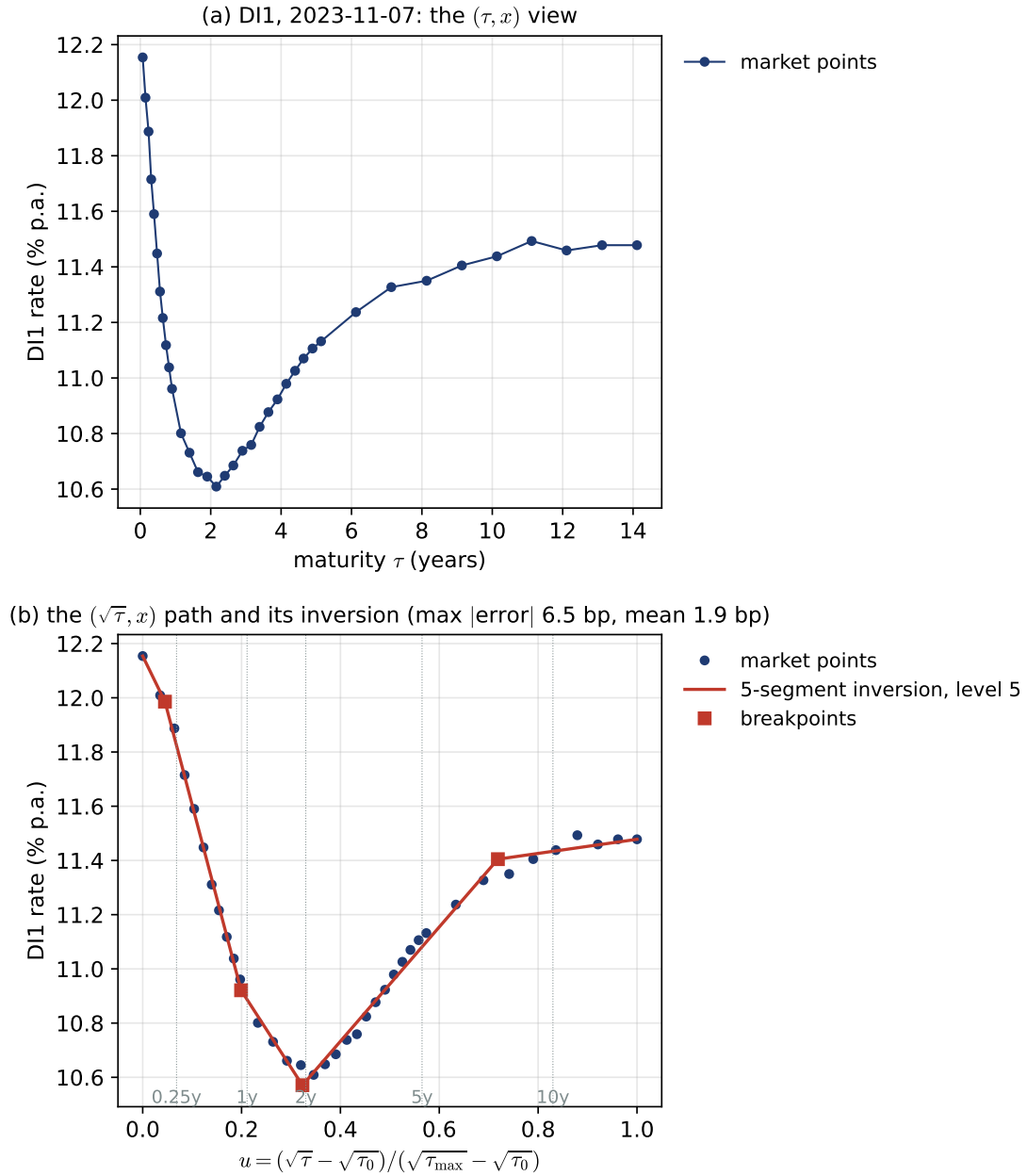


Figure 4: DI1 futures, 7 November 2023 (37 contracts, B3 settlement rates). (a) The curve against maturity. (b) The same curve on the $\sqrt{\tau}$ axis with its five-segment inversion at level 5: maximum error 6.5 bp, mean 1.9 bp at the market points; dotted lines mark maturities.

Figure 4 shows the curve and its inversion. The five segments place their four elbows at about 0.17, 0.91, 1.93 and 7.7 years ($u = 0.045, 0.199, 0.323, 0.719$). The five segments follow the features of the curve: a steep front end to about two months, a decline to about one year, the bottom of the trough between one and two years, a rise to nearly eight years, and a long end that is one straight segment in $\sqrt{\tau}$. The fit misses by at most 6.5 bp, at the 4–5 year convexity between the third and fourth elbows, and by 1.9 bp on average. With four segments instead of five the maximum error is 13–15 bp at level 5 and 10 bp at level 6, and the first elbow migrates to the front of the path: the fifth segment is what gives the front end its own piece.

5.2 The Treasury curve

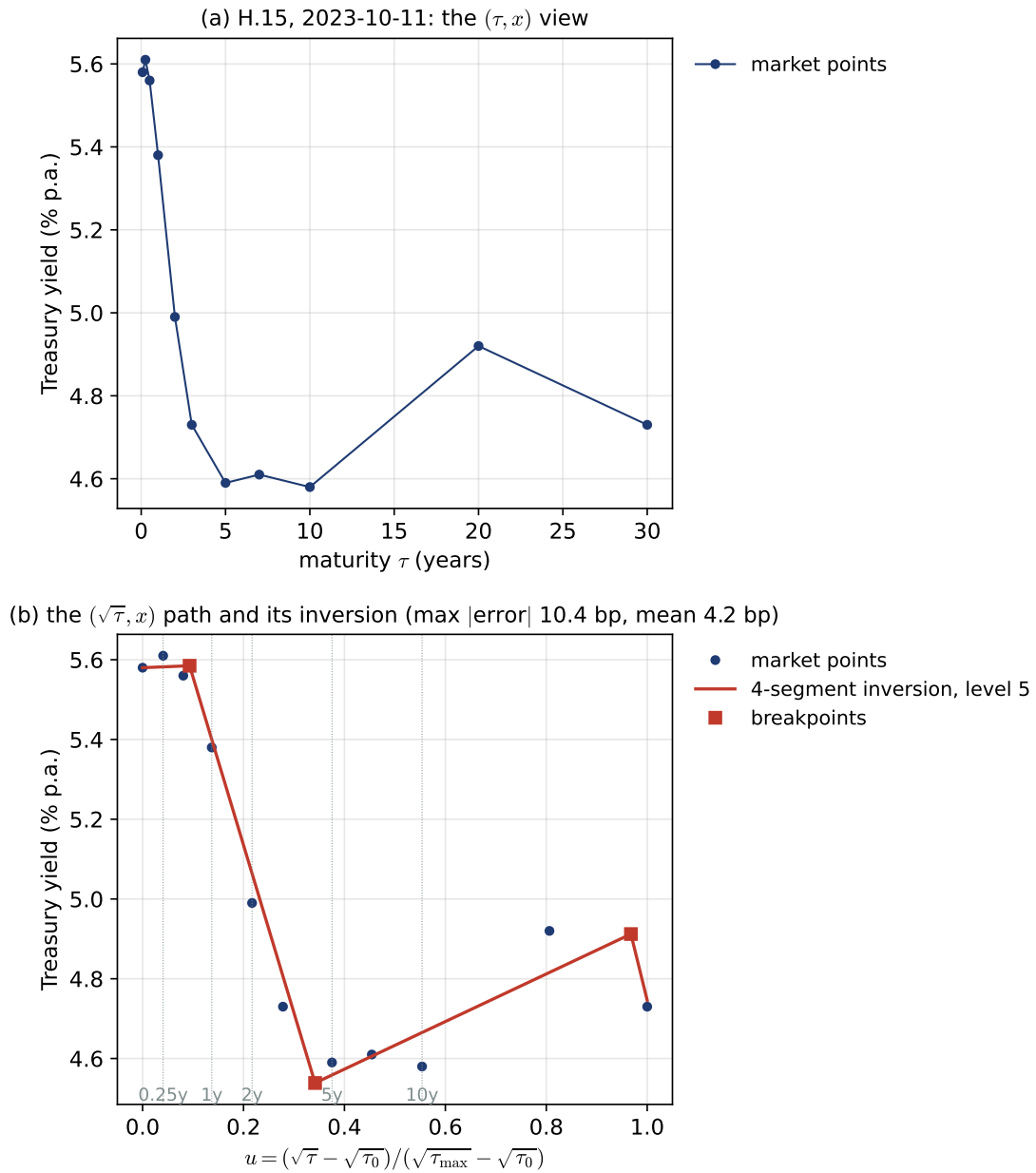


Figure 5: US Treasury constant-maturity yields, 11 October 2023 (eleven tenors, Federal Reserve H.15). (a) Against maturity. (b) On the $\sqrt{\tau}$ axis with the four-segment inversion at level 5: maximum error 10.4 bp, mean 4.2 bp.

The Treasury curve of that date is inverted at the front, flat between five and ten years, and humped at twenty years. Four segments at level 5 (Figure 5) place elbows at about 0.6, 4 and 28 years and fit to 10.4 bp maximum, 4.2 bp mean. The trough is located well; the twenty-year hump is located coarsely, the last elbow landing at 28 rather than 20 years and under-shooting the 20-year point by about 10

bp. Three segments give maximum errors of 24–27 bp at the tested levels. For four segments, the maximum error increases from 10.4 bp at level five to 23.0 bp at level six. The level-six magnitude weight is about three million times the level-one weight, so changing the truncation changes the relative emphasis of the objective strongly. The comparison says which truncation fits this curve better; it says nothing about what level six contains.

5.3 Levels, segments and the maturity axis

Curve	Date	n	m	max err (bp)	mean err (bp)	breakpoints u
DI1	2023-11-07	4	5	13.0	3.8	0.020, 0.273, 0.772
DI1	2023-11-07	4	6	10.2	3.5	0.273, 0.367, 0.727
DI1	2023-11-07	5	5	6.5	1.9	0.045, 0.199, 0.323, 0.719
DI1	2023-11-07	5	6	7.5	2.3	0.078, 0.170, 0.315, 0.722
H.15	2023-10-11	3	4	25.1	12.3	0.419, 0.980
H.15	2023-10-11	3	5	26.8	13.3	0.400, 0.980
H.15	2023-10-11	3	6	24.9	13.7	0.393, 0.926
H.15	2023-10-11	4	4	9.9	5.2	(square system)
H.15	2023-10-11	4	5	10.4	4.2	0.093, 0.342, 0.968
H.15	2023-10-11	4	6	23.0	13.5	0.383, 0.797, 0.817

Table 5: Fit error at the market points against segments n and level m , magnitude-normalised weights, segment-duration penalty threshold 2% of the path, 12–24 starts. Bold: the configurations shown in Figures 4 and 5. At level four there are seven nonconstant matching equations: fewer than the nine parameters for $n = 5$, and equal to the seven parameters for $n = 4$. These counts do not establish independence or solvability.

Table 5 is the sweep behind the two figures. It shows three things.

Levels. For the selected configurations of five DI1 segments and four Treasury segments, level five has smaller mean and maximum errors than level six. For four DI1 segments, however, level six improves on level five. A useful truncation therefore depends on the target, candidate class and weighting. This sweep supplies no rank calculation for a parameterised market-curve family and is not an empirical verification of Theorem 9.

Signature error and pointwise error. These are different objectives. For DI1 with four segments, a 24-start search found a lower signature residual than a 4-start search and a worse maximum pointwise error (14.6 against 10.0 bp). The five-segment configurations reported here fit better in path space. The table reports unweighted errors at the market points.

The maturity axis. On the DI1 curve the $\sqrt{\tau}$ axis compresses the range of the magnitude-normalised weights over levels 1–6 about forty times (2.8×10^6 under (τ, x) to 7.2×10^4); on the Treasury curve it does not (2.8×10^6 against 3.4×10^6), for this set of tenors and rates. The square root changes the path, its signature and the approximation class — a segment straight in $\sqrt{\tau}$ is not straight in τ — so the best attainable fits need not coincide; they can still be compared at common market maturities. All fitted examples use the $\sqrt{\tau}$ axis. The weight range is a scaling diagnostic, not a condition number of the regression Jacobian.

6 Conclusions

Core signatures make the classical Lyndon-coordinate description of a truncated signature explicit through a shuffle-elimination procedure and substitution rules. We used these coordinates to separate exact recovery, local identification and approximation by weighted regression. The two-segment criterion combines an algebraic compatibility condition with an admissible breakpoint; the moment coordinates determine polynomial paths and yield finite-level stability on bounded polynomial families. A rank condition gives local parameter recovery, with a path-norm estimate under an additional regularity assumption.

The stochastic and deterministic weighting questions have different answers. Additive bridge noise gives an explicit covariance matrix for the moment coordinates. The experimental magnitude schedule instead scales entries relative to the target. For all-level comparisons, undamped

factorial growth can produce infinite residuals; damping yields a metric inducing uniform convergence on an anchored Lipschitz class, but does not establish a Sobolev norm equivalence or a quantitative inverse rate. The linear-spline approximation barrier further separates signature depth from the expressive capacity of a candidate path class.

The DI1 and Treasury examples illustrate the numerical approximation problem, with maximum errors of 6.5 and 10.4 bp in the selected configurations. They show sensitivity to weighting, truncation and the maturity representation. Quantitative stability constants for nondegenerate piecewise-linear families and inverse-modulus bounds for the all-level metric remain questions for further work. The framework has been adopted in practitioner work on quantitative-trading futuretesting; see [Bloch (2024)]. We thank Sam Cohen (Oxford) for helpful discussions.

Code and data. The accompanying source package includes the core module, the scripts for the displayed numerical examples, input snapshots for the two curve dates, the digit-5 knots, and saved weight-comparison and configuration-sweep results. Its curve loaders default to the included snapshots. The project repository is <https://github.com/MarcosCarreira/CoreSignatures>; the files used for this version are its release `wilmott-r1-2026-09`, folder `2026 R1`. The core module uses NumPy and SciPy; the data and figure scripts additionally use pandas and Matplotlib. The routines named in the text (`coresig.py`) and the short scripts in `listings/` are the code behind the displayed examples; the figure scripts reproduce the figures from the included snapshots. The supplementary material (rules, closed forms, and the proofs of Proposition 1 and of the stability theorems) is submitted with the paper and is also in the release. The displayed empirical results are those of the companion runs; the formulas and the data handling were checked separately, and the package records those checks.

Use of AI assistance. The author used Anthropic’s Claude and OpenAI’s Codex to review the mathematical content of the paper, tighten the exposition, supply the citation chain through the Lyndon basis theorem, Poincaré–Birkhoff–Witt and Radford’s theorem that identifies the core variables with the Lyndon coordinates of $G^n(\mathbb{R}^d)$, and help write the proofs in the supplement, the regression and weights sections, the Python module and scripts, and the term-structure section; OpenAI’s Codex was used to check hypotheses, revise the recovery and stability statements, and check numerical consistency. All mathematical claims, the choice of which results to state as open problems, and the final form of the paper are the author’s responsibility.

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